

dependent on *hello.o* which, in turn, is dependent on *hello.c*. Therefore 'make' recompiles *hello.c* to create a new *hello.o* and hence rebuild the target executable 'prog1'. Note that *factorial.c* is not recompiled. Now consider that you have been given an arbitrary Makefile, which specifies a list of source files as well as the list of dependencies and also a list of changed source files; write a program to identify which of the objects need to be rebuilt and the order in which they have to be rebuilt. Let me give you a hint. Can you visualise the dependencies as a graph?

7. You are given a set R of ' n ' rectangles lying in a plane. You are asked to determine whether any two of the rectangles in the set R intersect. What is the time complexity of your solution?
8. You are given an array ' A ' of N integers such that $A[1] \geq A[2]$ and $A[n-1] \leq A[n]$. An element $A[i]$ is a local minimum if it is less than or equal to both of its neighbours. Formally, we denote an element $A[i]$ as a local minimum if $A[i] \leq A[i-1]$ and $A[i] \leq A[i+1]$. You are asked to write a program to find a local minimum in the given array. The obvious solution is to do a scan of the entire array ' A ' in $O(N)$ time. Can you design an algorithm that can find the local minimum in $\log(N)$ time?
9. You are given a directed graph of the roads in India, with each edge between two cities being associated with an independent safety probability $p(e)$. The safety of a path in the graph is the product of the safety probabilities of the edges in the path. You are asked to find the safest path to travel from Bengaluru to Delhi. Can you design an algorithm for this?
10. You are given a set ' X ' of N intervals on the real line. The intervals are represented by two arrays, $XL[1..N]$ and $XR[1..N]$, which represent the left and right end points of each of the N intervals. A subset Y of X is a set of intervals such that any real value that is contained in some interval in X is also contained in some interval in Y . In such a case, Y is called the tiling path or tiling cover of X . The size of a tiling cover is the number of intervals in Y , which is needed to cover X . You are asked to write a program to compute the smallest tiling cover for X . What is the time complexity of your approach?
11. As we all know, a palindrome is a string that reads the same forwards and backwards such as 'x', 'pop' or 'noon'. Any string can be broken into a sequence of palindromes. For instance, the string 'popdad' can be split into two palindromes 'pop' and 'dad', whereas a trivial splitting would be to split it into 6, such as "p", "o", "p", "d", "a", and "d". You are asked to write a program to split the given string into the minimum number of palindromic sub-strings. Can you find the minimum number of palindromic substrings into which you can split the string "bubba banana"?
12. You are given an input string $A[1..N]$. Of all the sub-strings of A , find the longest sub-string ' s ' such

that ' s ' appears both forward and backward in A . The forward and backward sub-strings should not overlap. For example, for the input string "SAWMWASING", the output should be "SAW". For the input string "DYNAMICPROGRAMANYTIMES", the output should be "YNAM".

13. Let's suppose you are given a weighted graph G and its minimum spanning tree T . Now you are told that the weight of one of the edges ' e ' in the graph G has been changed from w_1 to w_2 . How would you recompute the minimum spanning tree when:
 - a. w_1 is smaller than w_2 .
 - b. w_1 is greater than w_2 . While the brute force solution would be to re-run the MST computation algorithm for the graph ' G ', can you come up with a better approach?
14. You are given an array A of N positive integers without any duplicates. Write a program to find the smallest positive integer ' k ' such that k is not a member of A . Is it possible to come up with an $O(N)$ algorithm for this problem? If yes, give the algorithm. If not possible, explain why.
15. Stack is a data structure that supports LIFO (Last In First Out). Implement a modified version of Stack, which pops out the most frequently added item.

My 'must-read book' for this month

This month's must-read book suggestion comes from one of our readers, Shanmugavel, and is titled 'A Discipline of Programming' by Edsger W Dijkstra. This is a programming classic. Though this is an old book and some of the algorithms have been super-ceded by better and newer ones, it is still worth a read to understand the thinking behind designing clever algorithms. Thanks Shanmugavel, for your suggestion.

If you have a favourite programming book or article that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful so I can mention it in the column. This would help many readers who want to improve their software skills.

If you have any programming questions or software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming and here's wishing you the very best! 

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